

Notes: Goldsmith-Pinkham, Sorkin, and Swift (AER, 2020)

Bartik Instruments: What, When, Why, and How

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1 Big picture

Goldsmith-Pinkham, Sorkin, and Swift study the identification logic behind Bartik, or shift-share, instruments. A Bartik instrument is usually written as a weighted sum of national shocks, where the weights are local baseline shares. In the canonical labor-market case, the instrument predicts local labor-demand growth by interacting a city’s initial industry composition with national industry growth rates.

The paper’s central message is that a typical Bartik design should be read as a **pooled exposure design**. The baseline shares measure how exposed each location is to common shocks. Identification is therefore usually about the exogeneity of the **shares**, not simply about the national growth rates. The relevant question is: would locations with different baseline industry shares have had parallel outcome changes absent the common shocks?

The paper then opens the black box of the Bartik estimator. Since a Bartik instrument is a single weighted combination of many shares, the authors show how to compute **Rotemberg weights**: weights that reveal which individual share instruments drive the estimate and which identifying assumptions matter most. These weights can be interpreted as sensitivity-to-misspecification elasticities: if a high-weight share is endogenous, the Bartik estimate is especially vulnerable.

The paper’s practical contribution is a toolkit. A researcher using a Bartik instrument should report which shares receive high Rotemberg weight, examine whether those shares are correlated with observable confounders, test pretrends when possible, compare alternative estimators and overidentification tests, and visualize the dispersion of just-identified estimates.

2 Incremental contribution of the paper

1. **Clarifies identification in shift-share designs.** The paper gives a formal equivalence result: using the Bartik instrument in two-stage least squares is numerically equivalent to a GMM estimator that uses the underlying shares as instruments with a weight matrix built from the shocks.
2. **Reframes the usual identifying assumption.** In many empirical papers, researchers describe Bartik as differential exposure to common shocks. This description implies an identifying assumption on the shares: shares must not predict outcome changes except through the endogenous variable.
3. **Connects Bartik designs to difference-in-differences logic.** When outcomes are changes and shares are measured before the shock, the design asks whether high-exposure and low-exposure places would otherwise have followed parallel trends.
4. **Develops Rotemberg-weight diagnostics.** The paper shows how to decompose the Bartik estimate into a weighted average of just-identified IV estimates using each share separately. The largest weights identify the most important exposure comparisons.
5. **Warns about negative weights and heterogeneous effects.** If treatment effects differ across locations, negative Rotemberg weights can make the Bartik estimate hard to interpret as a convex average of causal effects.
6. **Provides a practical reporting checklist.** The paper turns a common instrument into an auditable research design: report top weights, assess covariate balance/correlates, test pretrends,

compare estimators, and visualize heterogeneity.

7. **Applies the framework to two influential examples.** The canonical labor-supply application looks fragile: a few industries drive the estimate, covariates are correlated with shares, and overidentification tests reject. The immigrant-enclave application looks more reassuring in estimates, but pretrends for college workers raise concerns.

3 Research question

The paper asks: *What does a Bartik instrument use for identification, when is that identification plausible, why do particular instruments receive influence, and how should researchers diagnose the design?*

This question matters because Bartik instruments are used widely in labor, regional, public, development, trade, macroeconomics, and finance. The instrument is easy to build because it combines accounting identities, but ease of construction does not guarantee a valid quasi-experiment.

4 Basic notation and setup

In the canonical cross-sectional labor-market example, the structural equation is

$$y_l = \rho + \beta_0 x_l + \epsilon_l,$$

where y_l is wage growth in location l , x_l is employment growth, and β_0 is the inverse labor-supply elasticity.

Employment growth can be decomposed using industry shares and local industry growth rates:

$$x_l = \sum_k z_{lk} g_{lk}.$$

The local industry growth rate can be decomposed into a national industry component and a local idiosyncratic component:

$$g_{lk} = g_k + \tilde{g}_{lk}.$$

The Bartik instrument is then

$$B_l = \sum_k z_{lk0} g_k,$$

where z_{lk0} is the baseline share of industry k in location l and g_k is the national industry growth rate.

In panel form, the instrument is

$$B_{lt} = \sum_k z_{lk0} g_{kt},$$

usually with baseline shares interacted with shocks in each time period and with location and time fixed effects or other controls.

5 Main formal results

5.1 Equivalence between Bartik IV and shares-as-instruments GMM

The paper’s first formal insight is a numerical equivalence. In the simple cross section, let Z be the matrix of industry shares and let G be the vector of national growth rates. Then $B = ZG$ is the Bartik instrument. The authors show that two-stage least squares using B is equivalent to a GMM estimator using all the columns of Z as instruments with the weight matrix

$$W = GG'.$$

This result is important because it changes how one interprets the instrument. Bartik is not just one mysterious generated shock. It is a particular way of combining many share instruments. The assumptions required for validity can therefore be discussed at the level of the underlying shares.

5.2 Identification under the shares view

A sufficient condition for consistency is strict exogeneity of the shares:

$$E[\epsilon_{lt} z_{lk0} \mid D_{lt}] = 0$$

for all relevant industries k , conditional on controls D_{lt} . In words, baseline exposure shares should not predict unobserved outcome changes except through the endogenous variable.

This is weaker than requiring shares to be unrelated to levels. Industry shares are equilibrium objects and may be strongly related to the level of wages, employment, education, or demographics. What matters for a typical design in changes is whether these shares predict *changes* in the structural error.

5.3 Shares view versus shocks view

The authors contrast their shares-based interpretation with the shocks-based interpretation emphasized by Borusyak, Hull, and Jaravel. Under the shocks view, consistency can come from many independent exogenous shocks even if the shares are not individually exogenous.

The paper argues that researchers should be explicit about which design they use. If the paper motivates the design by saying that some places are more exposed to particular industries or origin countries, or if it uses a two-industry intuition, it is usually appealing to the shares view. If the design relies on many random shocks averaging out bias in shares, it is appealing to the shocks view.

5.4 Rotemberg weights

The Bartik estimator can be decomposed as

$$\hat{\beta}_{Bartik} = \sum_k \hat{\alpha}_k \hat{\beta}_k,$$

where $\hat{\beta}_k$ is the just-identified IV estimate using share z_k as the instrument, and

$$\hat{\alpha}_k = \frac{g_k Z'_k X^\perp}{\sum_{k'} g_{k'} Z'_{k'} X^\perp}.$$

The weights sum to one, but they can be negative. A large $\hat{\alpha}_k$ means that the overall Bartik estimate is highly sensitive to the identifying assumption for that particular share.

This is the practical heart of the paper. Instead of defending every industry share equally, researchers should first ask which shares actually matter. The most important shares are the ones with high absolute Rotemberg weights.

5.5 Heterogeneous effects and negative weights

Under constant effects, negative Rotemberg weights are not conceptually problematic. With heterogeneous effects, they can become important. The paper considers a restricted form of location-specific heterogeneity and shows that even if each individual instrument has a reasonable local-average interpretation, the combined Bartik estimator can put nonconvex weights on underlying treatment effects.

The implication is practical: when high-weight instruments have very different just-identified estimates or when negative-weight instruments are outliers, the overall Bartik estimate may not have a simple treatment-effect interpretation.

6 Recommended diagnostic workflow

1. **State the identifying assumption.** Is identification coming from shares, shocks, or both? Do not leave this implicit.
2. **Report Rotemberg weights.** List the industries, origin countries, banks, or other exposure components with the largest positive and negative weights.
3. **Study correlates of high-weight shares.** If high-weight shares are correlated with demographics, industrial structure, trends, or other confounders, the exclusion restriction becomes harder to defend.
4. **Test pretrends when a pre-period exists.** A good exposure design should not show that baseline shares predicted outcome changes before the relevant shock.
5. **Compare alternative estimators.** TSLS with all shares, LIML, many-instrument robust estimators, and overidentification tests can reveal whether the single Bartik combination hides conflicting instruments.
6. **Visualize heterogeneity.** Plot the just-identified estimates $\hat{\beta}_k$ against first-stage strength, scaling points by Rotemberg weights and marking positive versus negative weights.

7 Application I: canonical labor-supply elasticity

The first application estimates the inverse labor-supply elasticity using U.S. Census data. The unit is a commuting zone. The outcome is earnings growth, the endogenous variable is employment

growth, and the Bartik instrument is constructed from 1980 industry shares and national industry growth rates across 1980-1990, 1990-2000, and 2000-2010.

There are 722 continental U.S. commuting zones and 228 nonmissing 3-digit IND1990 industries in 1980. Regressions are weighted by 1980 population. The paper emphasizes the leave-one-out construction of national industry growth rates to avoid using a location's own idiosyncratic growth in the national shock.

The empirical lesson from this application is cautionary. The Bartik estimate is heavily influenced by a few industries, especially oil and gas extraction and motor vehicles. These shares correlate with observable characteristics of places. Alternative estimators differ, and overidentification tests reject, suggesting that the identifying assumptions behind all the individual share instruments are not jointly convincing.

8 Application II: immigrant enclave instrument

The second application reinterprets the immigrant enclave instrument in Card (2009) as a Bartik-like design. The instrument interacts baseline location-origin immigrant shares with national inflows from origin countries. The outcome is the native-immigrant wage gap, and the endogenous variable is the relative immigrant-to-native labor supply.

The relevant shares are 1980 origin-country immigrant shares in cities. The shocks are origin-country immigrant inflows from 1990 to 2000. The paper analyzes high-school equivalent and college equivalent workers separately.

The high-school application is dominated by Mexico: the Mexico share receives almost half the Rotemberg weight. This is not necessarily bad, but it makes the design concrete: the key comparison is between cities with larger and smaller Mexican immigrant enclaves in 1980. Pretrends look relatively reassuring for high-school workers. For college workers, however, the pretrend figures are less reassuring, especially for the Philippines and other high-weight origins.

9 Main takeaways

The Bartik instrument is not a magic way to create exogenous variation. It is a weighted combination of many exposure designs. Its credibility depends on whether the high-weight exposures are plausibly exogenous to outcome changes.

The paper's most useful empirical prescription is simple: do not just report the Bartik coefficient. Report what drives it. A convincing Bartik paper should make the major exposure comparisons transparent and should show why those comparisons are credible.

The practical checklist is:

- explain whether identification relies on shares or shocks;
- report the top Rotemberg weights;
- inspect covariate balance or correlates of the high-weight shares;
- test pretrends whenever the data allow it;

- compare alternative estimators and overidentification tests;
- visualize just-identified estimates and negative weights.

10 Extracted figures and tables from the paper

10.1 Table 1—Summary of Rotemberg Weights: Canonical Setting

Read. Table 1 decomposes the canonical labor-supply Bartik estimate. Panel A shows that positive Rotemberg weights sum to 1.368 and negative weights sum to -0.368; in absolute-weight terms, negative weights account for about 21.2 percent. Panel D shows the five largest positive-weight industries: oil and gas extraction, motor vehicles, Other, guided missiles, and blast furnaces. Oil and gas extraction alone has weight 0.229.

Interpretation. This table shows that the canonical Bartik design is not broadly spread over hundreds of industries. A small number of industries drives a large portion of the estimate. The correlation between Rotemberg weights and national growth rates is nearly zero, so looking only at which industries grew or shrank nationally is a poor guide to the identifying variation. The weights are more correlated with cross-location variance in the shares, which means geographically concentrated industries receive influence. The top-weight industries therefore become the substantive research design: the estimate depends heavily on comparisons between places more and less exposed to oil and gas, automobiles, and a few other concentrated industries.

10.2 Table 2—Relationship between Industry Shares and Characteristics: Canonical Setting

Read. Table 2 regresses high-weight 1980 industry shares and the 1980-share Bartik instrument on commuting-zone characteristics. The R^2 values are large in several columns, especially Other (0.46) and the Bartik column (0.77). Several high-weight shares are correlated with demographics such as native-born share, education, veteran share, and number of children.

Interpretation. This table probes the exclusion restriction. If the Bartik design is a shares-based exposure design, then the shares should not predict unobserved changes in labor-supply conditions. The table does not prove invalidity, but it shows that high-weight shares are not random across places. For example, shares that are supposed to capture labor-demand exposure are correlated with variables related to labor supply and migration. That makes the research design less transparent and increases the burden of controls and explanation.

TABLE 1—SUMMARY OF ROTEMBERG WEIGHTS: CANONICAL SETTING

	Sum	Mean	Share		
<i>Panel A. Negative and positive weights</i>					
Negative	−0.368	−0.004	0.212		
Positive	1.368	0.010	0.788		
	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	$\text{var}(z_k)$
<i>Panel B. Correlations</i>					
$\hat{\alpha}_k$	1				
g_k	−0.015	1			
$\hat{\beta}_k$	0.017	−0.495	1		
$\hat{F}_k$	0.476	−0.032	0.016	1	
$\text{var}(z_k)$	0.549	−0.036	−0.003	0.316	1
	Sum	Mean			
<i>Panel C. Variation across years in $\hat{\alpha}_k$</i>					
1980	0.458	0.002			
1990	0.182	0.001			
2000	0.360	0.002			
	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95 percent CI	Ind. Share
<i>Panel D. Top five Rotemberg weight industries</i>					
Oil + gas extraction	0.229	0.034	1.170	(0.80, 1.90)	0.568
Motor vehicles	0.140	−0.017	1.525	(1.30, 1.90)	1.404
Other	0.091	−0.062	0.759	(0.10, 1.70)	1.697
Guided missiles	0.069	0.047	0.115	(−2.20, 0.70)	0.236
Blast furnaces	0.058	−0.078	1.084	(0.60, 5.10)	0.800
	α -weighted sum	Share of overall β	Mean		
<i>Panel E. Estimates of β_k for positive and negative weights</i>					
Negative	−0.074	−0.061	1.622		
Positive	1.290	1.061	−0.584		

Notes: This table reports statistics about the Rotemberg weights. In all cases, we report statistics about the aggregated weights with normalized growth rates, where we aggregate a given industry across years as discussed in Section IIIC and normalize growth rates to the per-period average as discussed in Section IIIB. Panel A reports the share and sum of negative weights. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of growth (g_k), the just-identified coefficient estimates ($\hat{\beta}_k$), the first-stage F -statistic of the industry share ($\hat{F}_k$), and the variation in the industry shares across locations ($\text{var}(z_k)$). Panel C reports variation in the weights across years. Panel D reports the top five industries according to the Rotemberg weights. The g_k is the national industry growth rate, $\hat{\beta}_k$ is the coefficient from the just-identified regression, the 95 percent confidence interval is the weak instrument robust confidence interval using the method from Chernozhukhov and Hansen (2008) over a range from −10 to 10, and *Ind. Share* is the industry share (multiplied by 100 for legibility). Panel E reports statistics about how the values of $\hat{\beta}_k$ vary with the positive and negative Rotemberg weights. The Other industry is the N/A code in the IND1990 classification system and includes full-time military personnel.

Table 1—Summary of Rotemberg Weights: Canonical Setting

TABLE 2—RELATIONSHIP BETWEEN INDUSTRY SHARES AND CHARACTERISTICS: CANONICAL SETTING

	Oil and gas extraction	Motor vehicles	Other	Guided missiles	Blast furnaces	Bartik (1980 shares)
Male	1.319 (0.242)	−0.501 (0.160)	4.076 (0.600)	0.126 (0.063)	0.344 (0.159)	−0.178 (0.035)
White	0.043 (0.102)	−0.714 (0.653)	−1.310 (0.281)	0.057 (0.043)	−0.681 (0.256)	−0.088 (0.029)
Native born	0.364 (0.092)	−0.129 (0.110)	0.824 (0.281)	−0.157 (0.133)	−0.312 (0.129)	−0.172 (0.019)
12th grade only	−1.096 (0.218)	1.283 (0.392)	1.040 (0.356)	−0.193 (0.091)	0.202 (0.150)	0.036 (0.030)
Some college	−0.311 (0.143)	0.687 (0.520)	1.060 (0.288)	0.033 (0.072)	−0.808 (0.254)	0.376 (0.042)
Veteran	−0.295 (0.227)	0.895 (0.917)	−5.793 (0.879)	0.202 (0.126)	2.526 (0.714)	0.000 (0.072)
Number of children	−0.043 (0.142)	0.954 (0.538)	−2.409 (0.558)	−0.006 (0.047)	0.003 (0.223)	−0.070 (0.034)
R^2	0.24	0.08	0.46	0.27	0.23	0.77
Observations	722	722	722	722	722	722

Notes: Each column reports results of a single regression of a 1980 industry share on 1980 characteristics. The final column is the Bartik instrument constructed using the growth rates from 1980 to 1990. Results are weighted by 1980 population. Standard errors in parentheses. The Other industry is the N/A code in the IND1990 classification system and includes full-time military personnel.

Table 2—Relationship between Industry Shares and Characteristics: Canonical Setting

10.3 Table 3—OLS and IV Estimates: Canonical Setting

Read. Table 3 reports OLS and IV estimates of the inverse labor-supply elasticity. OLS estimates are about 0.71 without controls and 0.63 with controls. TSLS using the Bartik instrument gives larger estimates, around 1.65 without controls and 1.22 with controls. The overidentified TSLS and MBTSLS estimates are much lower, around 0.67-0.76, while HFUL is much larger. Overidentification tests reject strongly for the overidentified specifications.

Interpretation. The table shows that the canonical Bartik result is sensitive to how the many instruments are combined. If all industry-share instruments were equally credible and estimated the same constant effect, alternative estimators should look more similar and overidentification tests should not reject so strongly. Instead, the Bartik row is not simply a robust summary of all shares. It is one particular weighted combination, and the disagreement across estimators suggests either misspecification, meaningful heterogeneity, or both.

10.4 Figure 1—Heterogeneity of $\hat{\beta}_k$: Canonical Setting

Read. Figure 1 plots each individual industry-share IV estimate $\hat{\beta}_k$ against its first-stage F -statistic. Circle size reflects the absolute Rotemberg weight, with circles for positive weights and diamonds for negative weights. The dashed horizontal line is the overall Bartik estimate with controls from Table 3.

Interpretation. The figure makes the overidentification problem visible. There is substantial dispersion across just-identified estimates even among instruments with reasonable first-stage strength. Some high-weight points are far from the overall estimate, and some outlying points have negative weights. If the researcher interprets the design through heterogeneous treatment effects rather than invalid instruments, the figure still raises a concern: the overall Bartik estimate may not be a simple convex average of economically meaningful local effects.

	Δ Emp		Coefficient equal	Over ID test
	(1)	(2)		
OLS	0.71 (0.06)	0.63 (0.07)	[0.04]	
TSLS (leave-out Bartik)	1.76 (0.33)	1.28 (0.42)	[0.23]	
TSLS (Bartik)	1.65 (0.34)	1.22 (0.15)	[0.19]	
TSLS	0.74 (0.05)	0.67 (0.07)	[0.10]	1,014.05 [0.00]
MBTSLS	0.76 (0.06)	0.69 (0.07)	[0.13]	
LIML	1.60 (0.00)	1.42 (0.57)	[0.76]	2,820.96 [0.00]
HFUL	2.85 (0.14)	2.69 (0.13)	[0.00]	804.19 [0.00]
Year and CZone FE	Yes	Yes		
Controls	No	Yes		
Observations	2,166	2,166		

Notes: This table reports a variety of estimates of the inverse elasticity of labor supply. The regressions are at the commuting zone level and the instruments are 3-digit industry-time periods (1980–1990, 1990–2000, and 2000–2010). Column 1 does not contain controls, while column 2 does. The *TSLS (Bartik)* row uses the Bartik instrument. The *TSLS* row uses each industry share (times time period) separately as instruments. The *MBTSLS* row uses the estimator of Anatolyev (2013) and Kolesár et al. (2015) with the same set of instruments. The *LIML* row shows estimates using the limited information maximum likelihood estimator with the same set of instruments. Finally, the *HFUL* row uses the HFUL estimator of Hausman et al. (2012) with the same set of instruments. The *J*-statistic for HFUL comes from Chao et al. (2014). The *p*-value for the equality of coefficients compares the adjacent columns with and without controls. The controls are the 1980 characteristics (interacted with time) displayed in Table 2. Results are weighted by 1980 population. Standard errors are in parentheses and are constructed by bootstrapping over commuting zones; a value of 0 in brackets indicates a value less than 0.05.

Table 3—OLS and IV Estimates: Canonical Setting

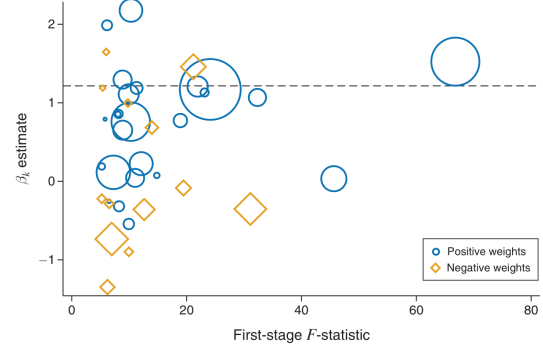


FIGURE 1. HETEROGENEITY OF $\hat{\beta}_k$: CANONICAL SETTING

This figure plots the relationship between each instruments' $\hat{\beta}_k$, first-stage F -statistics, and the Rotemberg weights.

Figure 1—Heterogeneity of $\hat{\beta}_k$: Canonical Setting

10.5 Table 4—Summary of Rotemberg Weights: Immigrant Enclave

Read. Table 4 reports Rotemberg weights for the immigrant enclave application. For high-school equivalent workers, Mexico receives weight 0.482, and the top five origins receive almost two-thirds of the weight. The correlation between the weights and the origin inflow shocks g_k is 0.991. For college equivalent workers, the top origin is the Philippines with weight 0.151, followed by Mexico, China, West Europe and others, and Cuba; the shock-weight correlation is 0.766.

Interpretation. This application differs from the canonical industry example. The shocks explain the weights much more strongly, and all weights are positive. But the weights also show that the design is highly concentrated. For high-school workers, the design is mostly a Mexico-enclave design. For college workers, it is more diffuse but still centered on a handful of origin countries. This makes the identifying assumptions concrete: one must believe that cities with larger 1980 Mexican or Filipino enclaves would not otherwise have had different wage-gap changes, conditional

on controls.

10.6 Table 5—Relationship between Origin Country Shares and Characteristics: Immigrant Enclave

Read. Table 5 regresses 1980 origin-country shares and the aggregate instruments on 1980 city characteristics. The R^2 values for the Bartik high-school and college instruments are 0.371 and 0.430, respectively. City size is positively related to several origin shares and to both aggregate instruments. Manufacturing share is not strongly related to most individual origin shares, except West Europe and others.

Interpretation. The table is the immigrant-enclave analogue of Table 2. The shares are not randomly assigned across cities; they are related to city size and other initial features. However, the covariate patterns appear less alarming than in the canonical industry application. The key point is that the table helps define the threat: if larger cities or cities with certain wage residuals were already on different trajectories, then baseline enclave shares could violate the exclusion restriction.

TABLE 4—SUMMARY OF ROTEMBERG WEIGHTS: IMMIGRANT ENCLAVE

Panel A. High school equivalent					
	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	$\text{var}(z_k)$
I. Correlations					
$\hat{\alpha}_k$	1				
g_k	0.991	1			
$\hat{\beta}_k$	0.169	0.164	1		
$\hat{F}_k$	0.203	0.173	0.181	1	
$\text{var}(z_k)$	0.043	-0.032	-0.106	-0.260	1
	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95 percent CI	
II. Top five Rotemberg weight origin countries					
Mexico	0.482	4.95e+06	-0.026	(-0.040, 0.000)	
El Salvador	0.054	4.65e+05	-0.046	(-0.070, -0.030)	
Philippines	0.050	5.31e+05	-0.023	(-0.040, 0.130)	
China	0.038	4.28e+05	-0.041	(-0.070, -0.010)	
West Europe and others	0.031	6.41e+05	-0.067	(-0.110, -0.050)	

Panel B. College equivalent					
	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	$\text{var}(z_k)$
I. Correlations					
$\hat{\alpha}_k$	1				
g_k	0.766	1			
$\hat{\beta}_k$	0.293	0.255	1		
$\hat{F}_k$	-0.028	-0.055	0.230	1	
$\text{var}(z_k)$	0.033	-0.381	-0.075	-0.225	1
	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95 percent CI	
II. Top five Rotemberg weight origin countries					
Philippines	0.151	6.32e+05	-0.065	(-0.125, -0.040)	
Mexico	0.102	5.44e+05	-0.062	(-0.095, 0.000)	
China	0.082	3.74e+05	-0.084	(-0.125, -0.060)	
West Europe and others	0.066	5.31e+05	-0.090	(-0.145, -0.065)	
Cuba	0.049	1.86e+05	-0.008	(-0.045, 0.500)	

Notes: This table reports statistics about the Rotemberg weights, which are all positive in this application. Panels AI and BI report correlations between the weights ($\hat{\alpha}_k$), the national component of growth (g_k), the just-identified coefficient estimates ($\hat{\beta}_k$), the first-stage F -statistics ($\hat{F}_k$), and the variation in the origin country shares across locations ($\text{var}(z_k)$). Panels AII and BII report the top five origin countries according to the Rotemberg weights. The *Others* are Australia, Cyprus, Israel, and New Zealand. The g_k is the number of immigrants from 1990 to 2000. $\hat{\beta}_k$ is the coefficient from the just-identified regression, the 95 percent confidence interval is the weak instrument robust confidence interval using the method from Chernozhukov and Hansen (2008) over a range from -10 to 10.

Table 4—Summary of Rotemberg Weights: Immigrant Enclave

TABLE 5—RELATIONSHIP BETWEEN ORIGIN COUNTRY SHARES AND CHARACTERISTICS: IMMIGRANT ENCLAVE

	Mexico	Philippines	El Salvador	China	Cuba	West Europe and others	Bartik high school	Bartik college
City size	0.054 (0.018)	0.026 (0.021)	0.106 (0.027)	0.057 (0.019)	0.049 (0.060)	0.039 (0.009)	0.059 (0.009)	0.023 (0.004)
College share	-0.545 (0.370)	0.559 (0.416)	0.692 (0.554)	1.318 (0.389)	-0.828 (1.206)	0.530 (0.175)	-0.021 (0.189)	0.157 (0.072)
Mean wage residuals for all natives	0.601 (0.388)	-0.428 (0.437)	0.595 (0.582)	-0.212 (0.408)	-0.199 (1.266)	0.052 (0.184)	0.267 (0.199)	0.041 (0.076)
Mean wage residuals for all immigrants	-0.652 (0.361)	0.596 (0.406)	-0.856 (0.540)	0.152 (0.379)	-0.079 (1.175)	-0.209 (0.170)	-0.385 (0.185)	-0.061 (0.070)
Mfg. share	0.059 (0.202)	-0.379 (0.228)	0.268 (0.303)	-0.192 (0.213)	-0.653 (0.660)	0.230 (0.096)	-0.006 (0.104)	-0.010 (0.039)
Observations	124	124	124	124	124	124	124	124
R^2	0.150	0.095	0.216	0.246	0.020	0.294	0.371	0.430

Notes: Each column reports results of a single regression of a 1980 origin country share on 1980 characteristics. Results are weighted by 1990 population. Standard errors in parentheses. For legibility, coefficients and standard errors of the first six columns are multiplied by 10,000,000. Coefficients and standard errors of the last two columns are not scaled. The *Others* are Australia, Cyprus, Israel, and New Zealand.

Table 5—Relationship between Origin Country Shares and Characteristics: Immigrant Enclave

10.7 Table 6—OLS and IV Estimates: Immigrant Enclave

Read. Table 6 reports OLS and IV estimates for high-school and college equivalent workers. For high-school workers, TSLS using the Bartik instrument gives about -0.04 with controls, and overidentified TSLS, MBTSLS, and LIML are very similar. For college workers, TSLS using Bartik gives about -0.08, and alternative estimators are again similar except HFUL, which has the opposite sign. Overidentification tests for TSLS do not reject in the displayed specifications, though HFUL-related tests reject.

Interpretation. Relative to the canonical labor-supply application, the immigrant-enclave estimates look more stable across estimators. This is reassuring because it means the individual origin-country instruments are not producing wildly different estimates. But stability of estimates is not enough. The pretrend figures remain crucial, especially for college workers, because serial correlation in immigrant inflows can make an enclave instrument capture ongoing location-specific trends rather than a clean shock.

10.8 Figure 2—Pretrends for High Rotemberg Weight Origin Countries: Immigrant Enclave, High School Equivalent

Read. Figure 2 plots reduced-form pretrend coefficients for the overall high-school instrument and the top high-school origin-country instruments: Mexico, El Salvador, Philippines, China, and West Europe and others. The 2000 point corresponds to the estimated reduced-form effect; earlier points assess whether the same shares predicted wage gaps before the main shock.

Interpretation. The high-school panels are relatively reassuring. The 1980 and 1990 coefficients are generally not large in the same way as the 2000 coefficients. The aggregate pattern resembles Mexico, which is expected because Mexico receives almost half of the Rotemberg weight. This supports the idea that the 1990s immigrant inflow generated a change rather than the 1980 shares merely capturing pre-existing wage-gap trends.

TABLE 6—OLS AND IV ESTIMATES: IMMIGRANT ENCLAVE

	Δ Emp		Coefficients equal (3)	Over ID test (4)
	(1)	(2)		
<i>Panel A. High school equivalent</i>				
OLS	-0.02 (0.01)	-0.03 (0.01)	[0.00]	
TSLS (Bartik)	-0.02 (0.01)	-0.04 (0.01)	[0.07]	
TSLS	-0.02 (0.01)	-0.04 (0.01)	[0.02]	43.30 [0.22]
MBTSLS	-0.03 (0.01)	-0.04 (0.01)	[0.08]	
LIML	-0.03 (0.01)	-0.04 (0.01)	[0.06]	73.16 [0.00]
HFUL	0.03 (0.01)	0.02 (0.00)	[0.26]	82.45 [0.00]
<i>Panel B. College equivalent</i>				
OLS	-0.06 (0.01)	-0.06 (0.01)	[0.65]	
TSLS (Bartik)	-0.08 (0.01)	-0.08 (0.01)	[0.93]	
TSLS	-0.06 (0.01)	-0.06 (0.01)	[0.71]	35.54 [0.54]
MBTSLS	-0.06 (0.01)	-0.07 (0.01)	[0.71]	
LIML	-0.06 (0.01)	-0.06 (0.01)	[0.72]	33.67 [0.63]
HFUL	0.04 (0.01)	0.04 (0.00)	[0.23]	67.95 [0.00]
Controls	No	Yes		
Observations	124	124		

Notes: This table reports a variety of estimates of the negative of the inverse elasticity of substitution between immigrants and natives. The regressions are at the city level and include a single time period (2000). The *TSLS* row is our replication of column 3 and column 7 of Table 6 in Card (2009). Column 1 does not contain controls, while column 2 does. The *TSLS (Bartik)* row uses the Bartik instrument. The *TSLS* row uses each origin country share separately as instruments. The *MBTSLS* row uses the estimator of Anatolyev (2013) and Kolesár et al. (2015) with the same set of instruments. The *LIML* row shows estimates using the limited information maximum likelihood estimator with the same set of instruments. Finally, the *HFUL* row uses the HFUL estimator of Hausman et al. (2012) with the same set of instruments. The *J*-statistic for HFUL comes from Chao et al. (2014). The *p*-value for the equality of coefficients compares the adjacent columns with and without controls. The controls are the contemporaneous characteristics displayed in Table 5. Results are weighted by 1990 population. Standard errors are in parentheses and are constructed by bootstrap over commuting zones. *p*-values are in brackets.

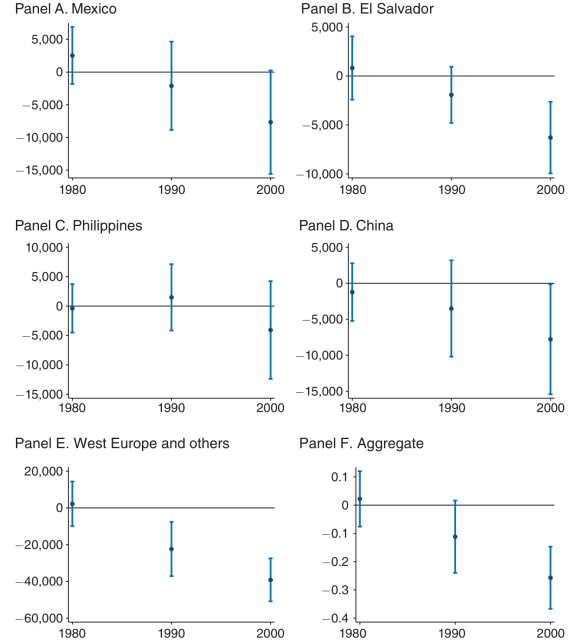


FIGURE 2. PRETRENDS FOR HIGH ROTEMBERG WEIGHT ORIGIN COUNTRIES:

Figure 2—Pretrends for High Rotemberg Weight Origin Countries: Immigrant Enclave, High School Equivalent

Table 6—OLS and IV Estimates: Immigrant Enclave

10.9 Figure 3—Pretrends for High Rotemberg Weight Origin Countries: Immigrant Enclave, College Equivalent

Read. Figure 3 repeats the pretrend exercise for college equivalent workers. The high-weight origins include the Philippines, Mexico, China, West Europe and others, and Cuba. Unlike the high-school panels, several of the college panels show negative coefficients already in 1980 and 1990, with magnitudes not clearly smaller than in 2000.

Interpretation. This is the main warning sign in the immigrant-enclave application. If 1980 enclave shares already predict wage gaps before the 1990-2000 inflow shock, then the design may be picking up persistent city-origin trends or serially correlated migration patterns rather than a clean contemporaneous supply shock. The authors connect this concern to Jaeger, Ruist, and Stuhler’s critique that immigrant inflows are often serially correlated.

10.10 Figure 4—Heterogeneity of $\hat{\beta}_k$: Immigrant Enclave

Read. Figure 4 plots individual origin-country just-identified estimates against first-stage strength for high-school and college equivalent workers. The dashed line is the overall Bartik estimate from Table 6. Point sizes reflect Rotemberg weights. Compared with Figure 1, the y-axis range is much

narrower and the high-weight points lie close to the overall estimate.

Interpretation. Figure 4 explains why the overidentified estimates in Table 6 are stable. Individual origin-country instruments do not generate the same degree of dispersion seen in the canonical industry application. For high-school workers, this visual evidence is reassuring when combined with the pretrend evidence. For college workers, the visual dispersion is not the main concern; instead, the concern comes from pretrends and the possibility that high-weight enclave shares track persistent inflow patterns.

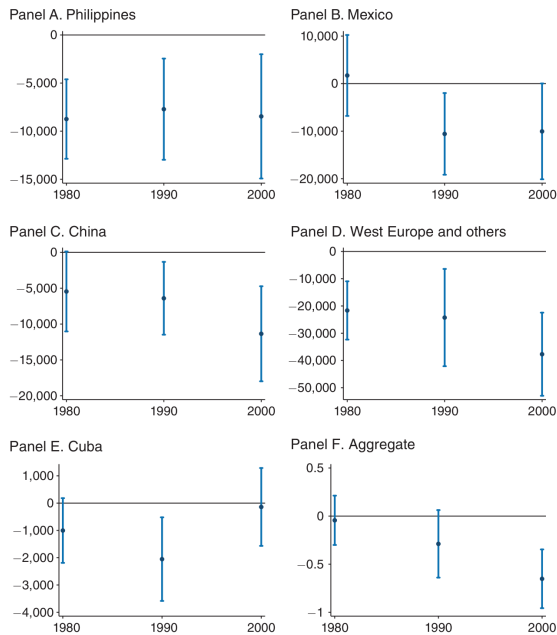


FIGURE 3. PRETRENDS FOR HIGH ROTEMBERG WEIGHT ORIGIN COUNTRIES: IMMIGRANT ENCLAVE, COLLEGE EQUIVALENT

Figure 3—Pretrends for High Rotemberg Weight Origin Countries: Immigrant Enclave, College Equivalent

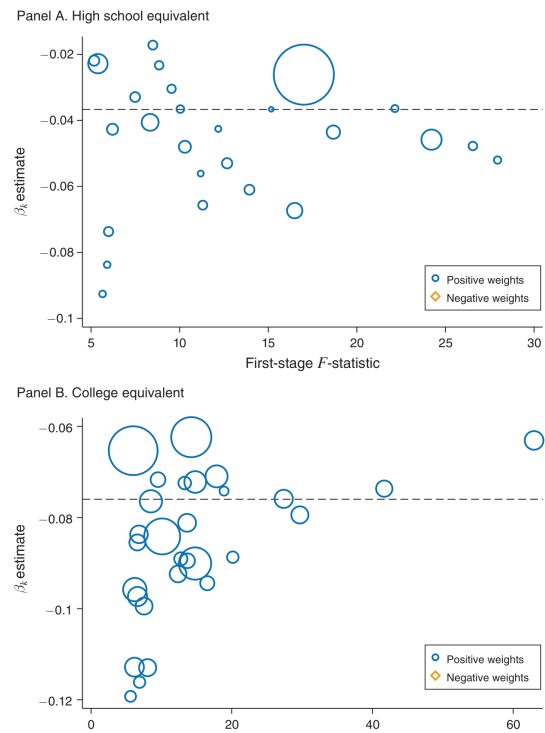


Figure 4—Heterogeneity of $\hat{\beta}_k$: Immigrant Enclave

11 How to use this paper in applied work

For any Bartik or shift-share empirical design, the paper suggests the following template:

1. **Define shares and shocks explicitly.** State what the shares are, when they are measured, and what the shocks represent.
2. **State the unit of causal variation.** In a shares design, the unit is the baseline exposure comparison, such as oil-intensive versus non-oil-intensive places or Mexico-enclave versus non-Mexico-enclave cities.
3. **Compute Rotemberg weights before interpreting the coefficient.** This reveals which exposure comparisons require the most credibility.
4. **Connect the top weights to economics.** If the largest weights are substantively plausible and pass diagnostics, the design becomes more credible. If the largest weights are surprising or

confounded, the design weakens.

5. **Treat pretrends as central, not optional.** A Bartik design in changes is a difference-in-differences-style design; pretrends are direct evidence about whether exposure shares predict changes before the shock.
6. **Do not overinterpret a single coefficient.** Alternative estimators, overidentification tests, and heterogeneity plots should shape how the coefficient is read.

12 Bottom line

The paper changes how Bartik instruments should be reported and defended. The main lesson is not that Bartik instruments are invalid. The lesson is that they are transparent only after researchers unpack them. A good Bartik application should show whose exposure matters, why those exposure shares are credible, and whether the estimate is robust to reasonable ways of combining the underlying instruments.